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Two-dimensional Nafion nanoweb anion-shield for improved electrochemical performances of lithium–sulfur batteries†

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We first demonstrate a facile and effective strategy to directly deposit a Nafion nanoweb onto a sulfur cathode to address the challenges underlying in Li–S batteries. The Nafion nanoweb not only blocks the diffusion of polysulfides by electrostatic repulsion but also allows fast Li-ion exchange with the sulfur cathode through the selective protection layer. The Nafion nanoweb-sulfur cathode retained an excellent cycle performance over 200 cycles and delivered an improved rate capability, as compared to a sulfur cathode and a Nafion layer-sulfur cathode.

Lithium–sulfur (Li–S) batteries are one of the promising candidates of next-generation batteries due to their various advantages such as a high theoretical capacity of 1672 mA h g^{-1}, environmental benignity, and cost-benefits.[1–4] Even though Li–S batteries can address several drawbacks of Li-ion batteries, some systemic challenges of Li–S batteries, associated with the polysulfide shuttling effect and low coulombic efficiency, still need to be overcome.

As an effective approach to confine polysulfides within the cathode side, the introduction of an interlayer in-between the sulfur cathode and the separator has been intensively investigated.[5–9] The Nafion polymer, as one of the promising interlayer materials, has shown great effectiveness in preventing the diffusion of polysulfides from the cathode electrode.[10–13] This is attributed to the presence of SO_3^- groups, which efficiently bounce the negatively charged polysulfide anions (S_n^{2-}) by electrostatic repulsion. Many studies have attempted to exploit this electrostatic property by (a) forming a Nafion layer on the sulfur cathode[12] or (b) modifying the pristine separator with the Nafion polymer.[10,11] However, the dense Nafion polymer layer that is formed on the electrode or separator impedes the efficient permeation of Li ions toward the sulfur cathode, thereby deteriorating the rate capability of the sulfur cathode.[10–12]

Herein, we first demonstrate a facile and effective strategy to directly deposit a Nafion nanoweb onto a sulfur cathode (a Nafion nanoweb-sulfur cathode) in order to efficiently suppress the polysulfide shuttling effect and enable fast Li-ion exchange with the sulfur cathode through the Nafion selective protection layer. The nanoweb structure is reported to have a great advantage in terms of its electrolyte wettability, which forms good ion migration channels and thus facilitates an efficient reaction between the active material and the electrolyte.[14] The deposition of the Nafion nanoweb was precisely controlled to have the same loading mass (0.84 mg cm^{-1}) as that of the Nafion film.[11] Due to the synergistic contributions of Nafion as a negatively charged polymer and the structural benefits of the nanoweb, the polysulfide shuttling was noticeably alleviated and the lithium ion permeability through the Nafion selective protection layer was dramatically improved, which led to a superior capacity and stable cycling performance at a high C-rate, as compared to those of a sulfur cathode and a Nafion layer-sulfur cathode.

Fig. 1a illustrates the process of directly depositing the Nafion nanoweb onto the sulfur cathode with the electro-spinning method and Fig. 1b depicts the role of the Nafion nanoweb as an ion-selective blocking layer in the sulfur cathode part. The Nafion nanoweb was prepared by electro-spinning the solution composed of a polyvinylpyrrolidone (PVP) polymer and a Nafion solution. PVP was used as a supporting polymer to sustain the architecture of the Nafion nanoweb. When lithium polysulfides diffused out of the electrode during discharge, the Nafion nanoweb that was formed on the sulfur electrode bounced the electrostatically negatively charged polysulfides by electrostatic repulsion; this led to the efficient suppression of the polysulfide shuttling effect. The main purposes of using the Nafion nanoweb are two-fold: (1) Nafion, possessing SO_3^- groups inside the chain, rejects the transfer of negatively charged polysulfides by

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† Electronic supplementary information (ESI) available: Preparation step of the sulfur cathode and Nafion nanoweb-sulfur cathode; EDS mapping of the sulfur cathode; cross-sectional image of the Nafion nanoweb-sulfur cathode; TEM images of PVP nanofibers and Nafion/PVP nanofibers; bar graph of average ζ potentials; and the SEM image of the Nafion layer-sulfur cathode. See DOI: 10.1039/c6ta04203a

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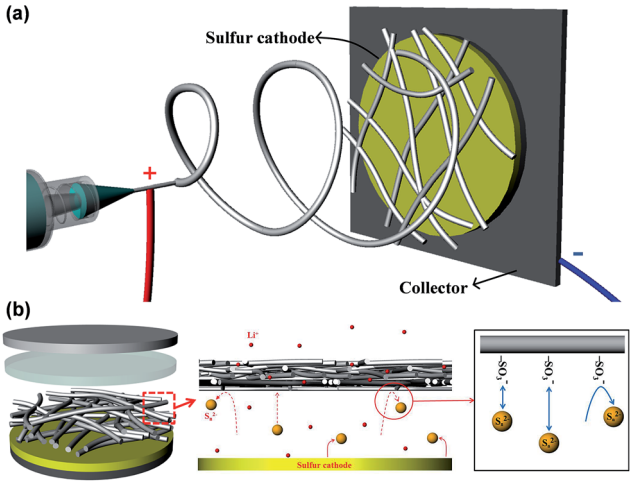

Fig. 1 (a) Schematic illustration showing deposition of the Nafion nanoweb onto the sulfur cathode and (b) the role of the Nafion nanoweb in the sulfur cathode during the discharge process.

Coulomb repulsion and (2) the Nanoweb structure allows efficient electrolyte wettability, which is associated with the rate capability, through the ion-selective blocking layer. These two synergistic contributions effectively alleviate the shuttling effect and facilitate fast electrochemical reaction between the sulfur cathode and Li ions, which result in outstanding electrochemical performances of Li–S batteries.

Fig. 2a shows a photographic image of a sulfur cathode and a Nafion nanoweb-sulfur cathode. The entire electrode turned white after depositing the Nafion nanoweb on the sulfur cathode, demonstrating the large-area deposition of a uniform Nafion nanoweb on the electrode. Fig. 2b presents the sulfur cathode, which is composed of sublimed sulfur, carbon black, and a binder. The carbon particles were homogeneously distributed throughout the electrode with no agglomerates. Fig. 2c clearly shows that the Nafion nanoweb, with a diameter of 1–2 μm, was densely formed on the sulfur cathode. To characterize the surface functionality of the Nafion nanoweb, Fourier-transform infrared spectrum (FT-IR) analysis was conducted; the results are presented in Fig. 2d. The peak at around 1050 cm^{-1} corresponds to the SO_3^- groups present in the polymer chain, which is clearly confirmed in both the Nafion layer and Nafion nanoweb.[15]

To ascertain the influence of Nafion on the net electrostatic charge in a polar solvent (DI water), the ζ potential was measured and is plotted in Fig. S4.† Fig. S1a† presents a bar graph of average ζ potentials for the PVP nanoweb, Nafion layer, and Nafion nanoweb. The PVP nanoweb itself exhibited a negligible electrical charge (near 0 mV) in the polar solvent, whereas the Nafion layer and Nafion nanoweb exhibited negative charges of −20.9 mV and −16.5 mV, respectively. The above results directly verified the influence of Nafion on the net electrical charges and provided evidence that the electrically negatively charged Nafion can bounce polysulfide anions by electrostatic repulsion.[11]

Fig. 3 presents the electrochemical performances of the sulfur cathode, the Nafion layer-sulfur cathode, and the Nafion nanoweb-sulfur cathode. The Nafion layer-sulfur cathode was prepared as a comparison sample in order to confirm the structural benefits of the nanoweb for the electrochemical performances. In Fig. S5,† the SEM image of the Nafion layer-sulfur cathode shows the dense and conformal coverage of the Nafion polymer over the whole area of the electrode. Fig. 3a shows the schematics of the Nafion layer-sulfur cathode and the Nafion nanoweb-sulfur cathode.

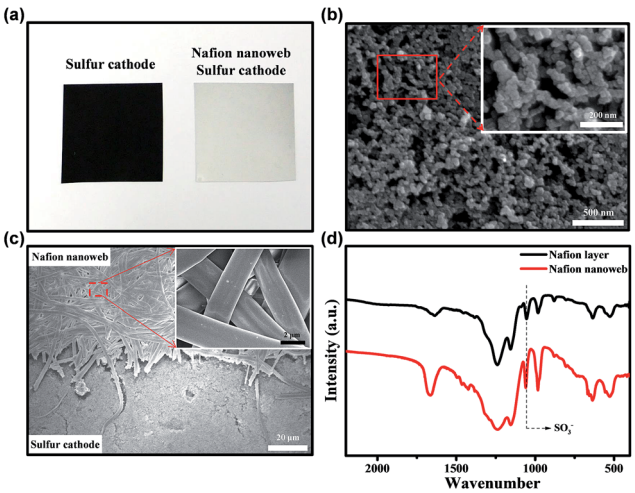

Fig. 2 Characterization: (a) photograph of the sulfur cathode and the Nafion nanoweb-sulfur cathode, SEM images of (b) the sulfur cathode and (c) the Nafion nanoweb-sulfur cathode, and (d) the FT-IR spectra.

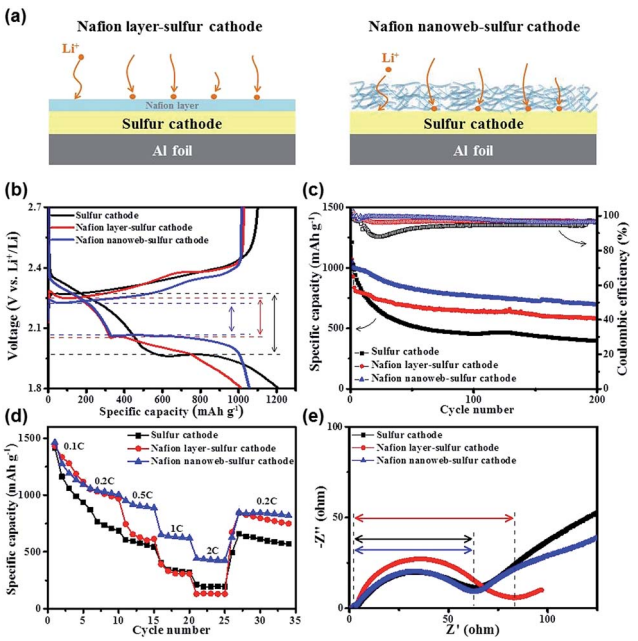

Fig. 3 Schematic illustrations and electrochemical performances of a Nafion layer-sulfur cathode and a Nafion nanoweb-sulfur cathode. (a) Schematics of each electrode, (b) initial discharge–charge curve, (c) cycle performance and coulombic efficiency, (d) Nyquist plots, and (e) rate capabilities (sulfur cathode, Nafion layer-sulfur cathode, and Nafion nanoweb-sulfur cathode).

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The specific capacity of the sulfur cathode, the Nafion layer-sulfur cathode, and the Nafion nanoweb-sulfur cathode at 0.5C was 1210 mA g^{-1}, 1016 mA g^{-1}, and 1056 mA g^{-1}, respectively. The initial specific capacities of the Nafion layer-sulfur cathode and the Nafion nanoweb-sulfur cathode were lower than that of the sulfur cathode; however, these results exhibited different properties in terms of the reversibility of the electrochemical reaction. Comparing the peak potential values between the lower plateau of the discharge and charge curves, the Nafion nanoweb-sulfur cathode showed the smallest gap, implying that the Nafion nanoweb can efficiently facilitate the reversible electrochemical reaction (Fig. 3b).[16]

The cycling performance and coulombic efficiency of each electrode were evaluated at a current density of 0.5C and are presented in Fig. 3c. In the initial cycle, the capacity of the sulfur cathode was relatively high, as compared with that of the Nafion layer-sulfur cathode and the Nafion nanoweb-sulfur cathode. However, the cycling retention of the sulfur cathode showed a fast decay during the first few cycles; this was due to the severe loss of the active material and the poor coulombic efficiency. After 200 cycles, the specific capacity reached 396 mA g^{-1} with a retention of 32.7%. With the aid of each Nafion selective protection layer (a Nafion layer and a Nafion nanoweb), the cycle performance and coulombic efficiency were dramatically improved. In particular, the Nafion nanoweb-sulfur cathode delivered a higher capacity of 695 mA g^{-1} with a retention of 65.8% over 200 cycles and also a higher coulombic efficiency of around 98% than the Nafion layer-sulfur cathode (582 mA g^{-1}, 54.3%). In terms of the cycling retention after electrochemical stabilization of the electrodes (after three cycles), the Nafion layer-sulfur cathode showed a comparable cycle retention performance but a lower capacity relative to the Nafion nanoweb-sulfur cathode (Fig. S7†). The two main effects (*i.e.*, efficient electrolyte permeability and Coulomb interaction) of the Nafion nanoweb synergistically facilitated better electrochemical reactions between the sulfur cathode and the electrolyte, thus outperforming the Nafion layer-sulfur cathode.

To clearly investigate the structural benefits of the one-dimensional nanoweb in terms of the rate capability, each electrode was evaluated at different current densities from 0.1C to 2C (Fig. 3d). In an initial cycle of each electrode, the specific capacities at 0.1C were almost identical. However, as the current density was increased from 0.2C to 2C, each electrode exhibited a different rate capability. For the sulfur cathode, the specific capacity abruptly decreased due to the harsh diffusion of polysulfides out of the sulfur cathode. The Nafion layer-sulfur cathode showed a better rate discharge capacity than the sulfur cathode and a similar rate capability to the Nafion nanoweb-sulfur cathode at current densities from 0.1C to 0.2C; however, the Nafion layer-sulfur cathode showed an inferior rate capacity from 0.5C to 2C because the dense Nafion layer hindered rapid Li-ion exchange with the sulfur cathode, thereby deteriorating the rate capability of the electrode.

The structural benefit of the nanoweb architecture, in terms of surface reactions, can be further confirmed by carrying out electrochemical impedance spectroscopy (EIS) studies in a frequency range of 100 kHz to 100 mHz with an amplitude of 10 mV (Fig. 3e). The ohmic resistances for the sulfur cathode, the Nafion layer-sulfur cathode, and the Nafion nanoweb-sulfur cathode were similar values: 1.27 Ω, 1.00 Ω, and 1.09 Ω, respectively. In contrast, the charge transfer resistances exhibited different values. For the Nafion layer-sulfur cathode, the charge resistance (82.7 Ω) was higher than that of the sulfur cathode (62.79 Ω) and the Nafion nanoweb-sulfur cathode (62.67 Ω). This can be due to the relatively poor electrolyte wettability of the dense Nafion layer relative to that of the Nafion nanoweb. In order to verify the structural benefit of the Nafion nanoweb structure, the wettability properties were confirmed by performing contact angle measurements of the electrolyte on the surface of each electrode (Fig. S8†). The contact angle of the electrolyte on the Nafion layer was 15.6°. In contrast, the electrolyte spread out on the surface of the Nafion nanoweb without forming a droplet. This directly demonstrated that the Nanoweb provides better wettability with the electrolyte compared to the dense layer.

To visually observe the effect that the Nafion nanoweb has on suppressing the migration of polysulfides, polysulfide diffusion tests were performed with a diffusion test bottle (Fig. 4). The diffusion test bottle was composed of a polysulfide solution, separator, and fresh electrolyte (Fig. 4a). The polysulfide solution was prepared by following the report of Dr Cui.[17] Instead of using a sulfur cathode electrode, three types of separators were prepared and inserted into the bottle: (1) a pristine separator, (2) a PVP nanoweb-deposited separator, and (3) a Nafion nanoweb-deposited separator. The deposition conditions are the same as those used while preparing the Nafion nanoweb-sulfur cathode.

As shown in Fig. 4b, the three diffusion test bottles were introduced into the fresh electrolyte to determine the diffusion rate of polysulfides through each separator. For the pristine separator, the polysulfides started to diffuse out of the bottle as soon as the bottle was dipped into the fresh electrolyte; this was

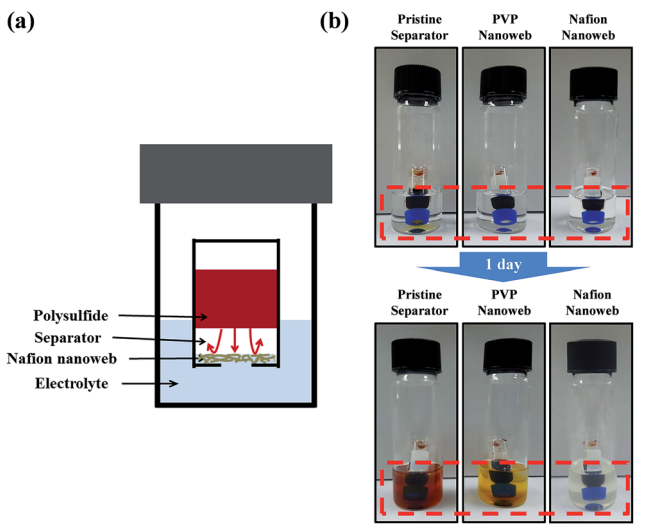

Fig. 4 (a) Configuration of a diffusion test bottle and (b) diffusion rate tests with a pristine separator, a PVP nanoweb-deposited separator, and a Nafion nanoweb-deposited separator.

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due to the concentration gradient. In contrast, the polysulfides slowly diffused out of the bottle with the PVP nanoweb-deposited separator and the Nafion nanoweb-deposited separator. After one day, the three diffusion test bottles showed marked differences in color. For the diffusion test bottle with the pristine separator, the color of the fresh electrolyte changed to dark brown, indicating the large migration of polysulfides out of the separator. In contrast, the color of the fresh electrolyte became light yellow when the Nafion nanoweb-deposited separator was applied, showing a significant suppression of polysulfide diffusion out of the selective protection layer. The PVP nanoweb-deposited separator also appeared to block the migration of polysulfides, but this effect was much less than that of the Nafion nanoweb-deposited separator. These results directly showed that the Nafion nanoweb can inhibit the diffusion of polysulfides in Li–S cells.

In summary, we first demonstrated a facile and efficient methodology to deposit a Nafion nanoweb onto a sulfur cathode in order to diminish the polysulfide shuttling effect and increase the coulombic efficiency. Compared to a sulfur cathode, the Nafion nanoweb-sulfur cathode retained more stable cycling performances over 200 cycles with high coulombic efficiency and delivered an improved rate capability. The excellent electrochemical performances of the Nafion nanoweb-sulfur cathode were attributed to the synergistic effect of the negatively charged Nafion polymer and the structural benefits of the nanoweb architecture. Unlike the previous concept of modifying the separator or the electrode with a Nafion solution, our proposed concept provides a new strategy to address the inherent drawbacks of Li–S batteries with a simple methodology. Additionally, this method can be further extended to other types of functional polymers (conducting polymer, hydrophilic polymers, *etc.*) or metal oxide-based materials.

Acknowledgements

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